



Enhancing keypoint detection in Y-Maze behavior test automation: introducing stadium heatmap and squared adaptive wing loss

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Abstract

The Y-Maze behavioral test evaluates short-term spatial learning and memory in experimental mice, assessing the impact of various substances or conditions. Traditional methods require continuous manual monitoring, which is labor-intensive and prone to observer fatigue. To streamline this process, we propose an automated approach that not only tracks the critical positions of the mice but also monitors the layout of the Y-Maze itself. We explored the efficacy of conventional General Gaussian heatmaps alongside our novel Ellipse and Stadium heatmaps. Our findings indicate that our Stadium heatmap significantly enhances the precision of position recognition within the maze compared to traditional methods. Additionally, we address a common issue in heatmap regression—the Adaptive Wing loss, where the first derivative can become excessively high. By introducing and testing the Squared Adaptive Wing loss, we demonstrate its ability to improve the reliability of our automated system.

Keywords Deep learning · Computer vision · Keypoint detection · Y-maze behavior test · Stadium heatmap · Squared adaptive wing loss

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1 Introduction

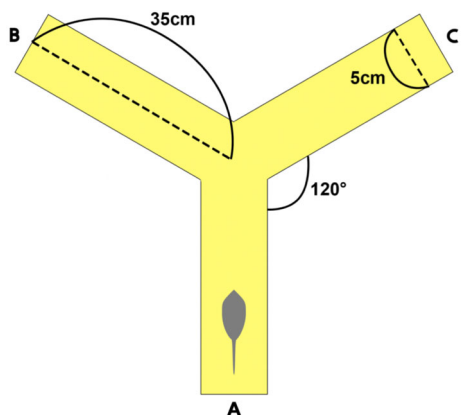
Precise preclinical analysis of short-term spatial learning and memory is essential for research and clinical trials on brain diseases like Alzheimer's. Behavioral tests are employed for this purpose, with the Y-Maze being a representative testing method [1]. Utilizing the characteristic of rodents to explore new environments, the Y-Maze test observes their behavior in a maze shaped like the letter 'Y', as shown in Fig. 1 [2]. This method can determine the effects of variables, such as drug administration, on memory and motivation. For example, we can evaluate the cognitive function improvement effect of DTMB drug in an Alzheimer's experimental model through the Y-maze experiment. Compared to the normal test group, the alternation (%) of the Alzheimer's group was significantly reduced, and cognitive decline was observed. In addition, in the case of the experimental group administered the DTMB drug, the alternation increased compared to the experimental group suffering from Alzheimer's disease, confirming the cognitive function improvement effect of the DTMB drug [3]. Therefore, in this experiment, it is important to observe behavioral patterns based on the location of the mice.

The Y-Maze experimental setup is designed to continuously generate data, necessitating an automated system to monitor and record the positions of experimental mice effectively. Manual tracking by human observers of such dynamic data is not only prone to errors but also requires significant labor. By implementing a precise prediction model for mouse positions, it is possible to free up time for observing the Y-maze.

In this context, keypoint detection techniques are employed to enhance accuracy. These techniques are akin to those used in Human Pose Estimation [4–7] and Face Alignment [8–11], aimed at identifying the precise locations of specific objects within the maze. One of the primary methods used is heatmap regression [9–11], which transforms the actual coordinates of each keypoint into a Gaussian heatmap. This method uses loss functions such as the Adaptive Wing loss [11] to produce a heatmap that closely resembles the induced feature map, thereby accurately estimating the central position of each keypoint. Moreover, Ellipse heatmaps [10] are also widely utilized, providing a viable alternative to the traditional Gaussian approach.

To enhance the analysis of Y-maze results and track the positions of experimental mice accurately, it is essential not only to detect keypoints such as the nose, body center, and tail's starting point of the mouse but also to monitor key reference points of the Y-maze

Fig. 1 A Y-maze comprising three arms A, B, and C. Each arm is spaced 120 degrees apart, and the length is 35 centimeters from the intersection of the three arms. The width of the arm is 5 centimeters



apparatus, including the center and endpoints of arms A, B, and C. Precise identification of these reference points is critical. In this study, we employed both the Ellipse and the newly introduced Stadium heatmaps to improve accuracy in identifying these points. The Stadium heatmap was explicitly used to pinpoint the endpoints of the arms, while the general Gaussian heatmap was used to locate the positions of the mice and the center of the apparatus. However, there has been no prior research on observation techniques for the Y-Maze behavior test, which led us to adopt and apply these innovative methods. We have incorporated commonly used benchmarks in this field, such as the HourglassNet architecture, heatmap regression methodology, and AdaptiveWing Loss, to ensure a comprehensive approach.

According to our findings, employing the Adaptive Wing loss with the general Gaussian heatmap for the arms' endpoints resulted in an average Normalized Mean Error (NME) of 0.88%. This metric, which measures the error based on the distance from the center of the apparatus to the arm endpoints, was reduced to 0.84% with the Ellipse heatmap and further to 0.85% with the Stadium heatmap. Additionally, the NME for each endpoint of arms A, B, and C were as follows: 0.73%, 0.69%, and 0.87% with the general Gaussian heatmap; 0.69%, 0.56%, and 0.72% with the Ellipse heatmap; and 0.72%, 0.62%, and 0.71% with the Stadium heatmap, respectively.

Our experiments highlighted a significant performance improvement when using the Stadium heatmap for the endpoints of arms A, B, and C, thereby enhancing the overall NME. While the Adaptive Wing loss is a common choice in heatmap regression methods, we observed that the first derivative values were excessively high at keypoints. To resolve this, we developed an optimized version termed the Squared Adaptive Wing loss. This modification led to a further reduction in the NME values to 0.77% (previously 0.88%) using the general Gaussian heatmap, 0.74% (previously 0.84%) with the Ellipse heatmap, and 0.72% (previously 0.85%) with the Stadium heatmap. Consequently, the Squared Adaptive Wing loss demonstrated superior performance improvements compared to the conventional Adaptive Wing loss.

Section 2, we cover the fundamentals of keypoint detection, including techniques such as heatmap regression, the application of four-stacked HourglassNet, and the Adaptive Wing loss method. Following this, Section 3 details the Squared Adaptive Wing loss, designed to overcome the limitations of the Adaptive Wing loss. This section also introduces the Ellipse and Stadium heatmaps, which are particularly effective for tracking the endpoints of the arms in the Y-maze apparatus.

Section 4 is dedicated to the implementation aspects of our study, discussing the dataset used, data augmentation methods, performance metrics, hyperparameters, and specifics of neural network training. Section 5 presents the results of our experiments, providing a comprehensive analysis of the findings. The concluding section summarizes the research and outlines potential future directions for further investigation.

The contributions of this paper can be summarized as follows:

1. We demonstrate the feasibility of implementing Y-Maze behavior test automation using Keypoint detection.
2. We provide fundamental guidelines for automating the Y-maze behavior test, including neural network structures, data configurations, and loss functions.
3. We propose the application of Ellipse and Stadium heatmaps for predicting the endpoints of Arms A, B, and C in the Y-Maze device and show that this enhances the prediction performance.

4. We analytically highlight the issues inherent in the Adaptive Wing loss function and introduce the Squared Adaptive Wing loss as an alternative solution, demonstrating an overall improvement in performance.

2 Keypoint detection

In Pose Estimation and Facial Alignment, heatmap regression is a prevalent method utilized in deep learning to identify locations. This technique converts the probability of coordinate presence into a heatmap format, and many high-performance studies have been built on this methodology. Noteworthy contributions to Facial Alignment research include STAR [8], HIH [9], ADNet [10], and AWing [11]. In the area of Pose Estimation, significant works such as UDP Pose [12], PSA [4], CCM [5], HRNet [6], and OmniPose [7] have shown impressive outcomes. An alternative strategy, the Coordinate regression method [13, 14], trains the neural network to output coordinates directly, with implementations like AnchorFace [13] and the Soft-argmax Layer [14] exemplifying this approach.

However, due to its enhanced performance and simplicity in training and analysis, we have chosen to implement heatmap regression. This paper discusses the necessary methodologies for accurately monitoring the reference positions within the Y-Maze setup and the locations of the experimental mice, as well as for evaluating the experimental outcomes.

2.1 Dataset and processing

As illustrated in Fig. 2, a heatmap is typically represented using a Gaussian distribution. This format highlights the keypoint within an image by marking it with the highest pixel value. Originally, keypoints were simple coordinates, but these were transformed into this heatmap format for processing. The neural network uses a loss function during training to refine its output, ensuring that the feature map it generates closely resembles the intended heatmap. During the inference stage, it is crucial to revert these heatmap values back to standard coordinates to interpret the model's findings accurately. For this conversion, we employ the

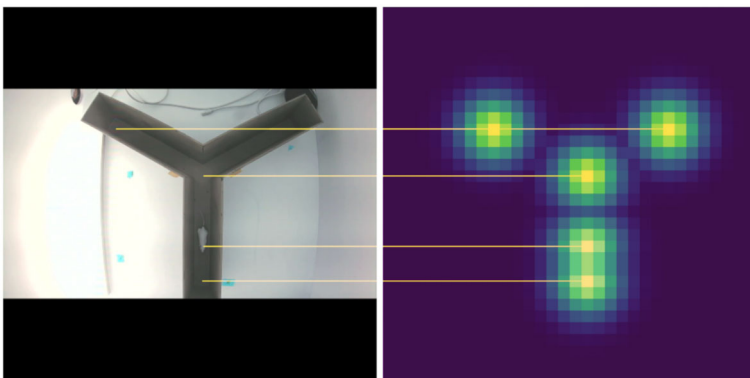


Fig. 2 Coordinates of each keypoint(left) and its heatmap represented by Gaussian distribution(right)

Soft-argmax function, detailed in (1), which effectively recalculates the coordinates from the heatmap data.

$$\text{Soft-argmax}(X) = \sum_i \frac{e^{x_i}}{\sum_j e^{x_j}} i \quad (1)$$

2.2 Architecture

HourglassNet [15], as shown in Fig. 3, is a neural network architecture widely used to achieve high performance in tasks such as Pose Estimation and Face Alignment. This network features an encoder-decoder structure similar to Unet [16], but it uniquely integrates internal connections within the encoder and decoder through a Residual Module, rather than simple skip connections. Each Hourglass Module in the network is effectively connected or stacked together, using the output from each module to compute the loss simultaneously. This setup promotes efficient backpropagation through the earlier layers, akin to the role of auxiliary classifiers in Inception [17]. This method of enhancing learning and precision is known as intermediate supervision.

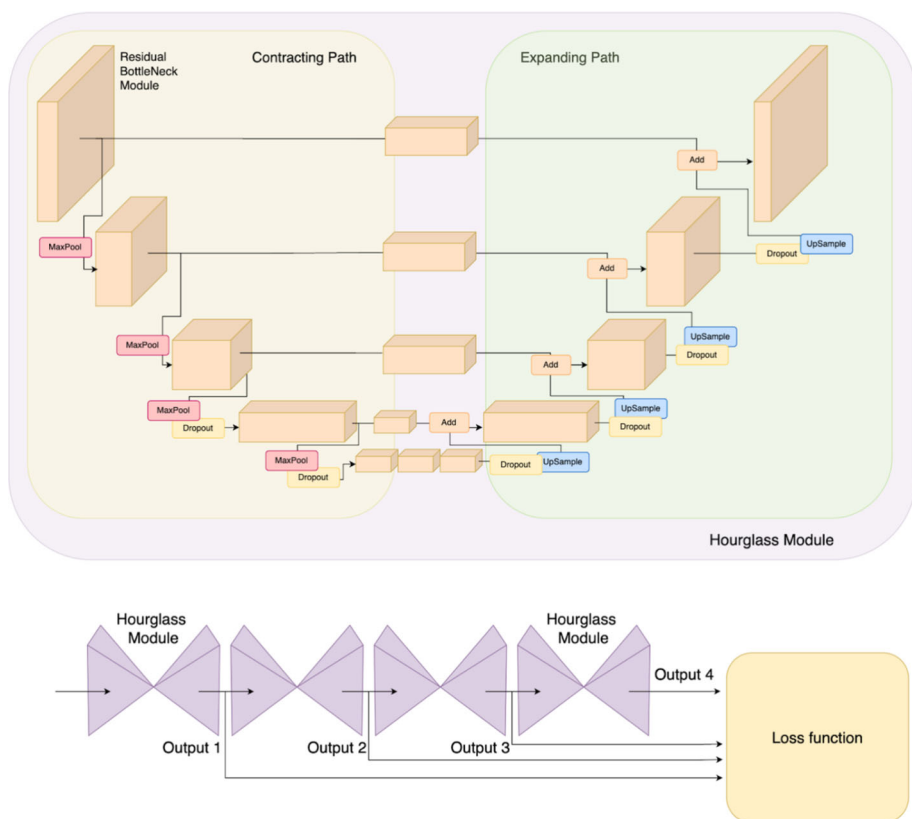


Fig. 3 Four-stacked HourglassNet architecture. It generates outputs from each hourglass module and computes the losses for backpropagation from the intermediate supervision point

This architecture offers the distinct advantage of simultaneously learning and evaluating connections across both shallow and deep layers. This capability allows for comprehensive integration of features at different levels, which is why it is extensively employed in various research studies.

2.3 Loss function

When transformed from coordinates, the heatmap delineates a localized area, often leading to imbalanced data, with the background substantially outnumbering the foreground. This imbalance can skew the training process, leading to a propensity for predicting the background or reducing overall model performance. To counter this, the Adaptive Wing loss, which utilizes adaptive weights to manage imbalanced data, is commonly used in high-performance models, particularly in Face Alignment tasks.

The mechanics of the Adaptive Wing loss are detailed from (2) to (4) where parameters are set as follows: $\alpha = 2.1$, $\omega = 14$, and $\theta = 0.5$. (2) and (3) compute the slope (A) and the y-intercept (C) for the linear scenario depicted in the 'otherwise' condition of (4). This setup ensures a smooth and continuous transition between the computed values across different scenarios in (4). The loss is tailored based on the actual value of Ground truth Y : if $Y = 1$, an exponent of 1.1 enhances the gradient; if $Y = 0$, an exponent of 2.1 reduces it. This feature allows the neural network to prioritize learning from locations with high Ground truth values, which optimally supports coordinate transformation using the Soft-argmax function that focuses on maximizing value deviations. As a result, most studies employing heatmap regression opt for the Adaptive Wing loss or its variants to ensure accuracy and efficiency.

$$A = \frac{\omega \cdot (\alpha - Y) \cdot \theta^{\alpha-Y-1}}{1 + \theta^{\alpha-Y}} \quad (2)$$

$$C = A \cdot \theta - \omega \cdot \log(1 + \theta^{\alpha-Y}) \quad (3)$$

$$\text{Adaptive Wing Loss}(Y, \hat{Y}) =$$

$$\begin{cases} \omega \cdot \log(1 + |Y - \hat{Y}|^{\alpha-Y}) & , |Y - \hat{Y}| < \theta \\ A \cdot |Y - \hat{Y}| - C & , \text{otherwise} \end{cases} \quad (4)$$

3 Proposed methods

The approach detailed in Section 2 ensures that the training and implementation of the Y-Maze keypoint detection would encounter no substantial challenges. However, our analysis has identified two critical areas that could benefit from further enhancements. Firstly, there is a notable concern with the Adaptive Wing loss function where its first derivative value remains excessively high. This could potentially impact the accuracy of the model under certain conditions. Secondly, the current shape of the heatmap does not accurately represent the specific contours of the Y-Maze arm ends, which could lead to less precise keypoint localization. Addressing these issues will refine our model and improve the overall performance.

3.1 Loss functions that converge to zero in the first derivative

As illustrated in Fig. 4, the gradient of the Adaptive Wing loss remains high as the loss nears zero and then decreases sharply. According to (4), the term $\alpha - Y$ in $|Y - \hat{Y}|^{\alpha - Y}$ is never less than 1.1, suggesting that theoretically, the gradient should also approach zero as the loss does. However, in practice, when using Adaptive Wing loss, the training loss often levels off around 0.0005, with a corresponding gradient near 7.2. Even as the training loss diminishes further to 0.0001 and 0.00001, the gradients are observed at approximately 6.3 and 4.87, respectively. This indicates that high gradients can still occur even when predictions are nearly perfect, potentially leading to unnecessarily large weight updates. Consequently, this high gradient condition persists near the loss minima, causing fluctuations in performance where it might significantly drop and then recover.

To mitigate the issue of high gradients and promote better convergence towards the optima, we have developed the Squared Adaptive Wing loss, as defined by (5)–(9). To enable a more gradual reduction in the gradient, we have chosen to utilize squared error instead of absolute error, as detailed in (9). To improve model performance, it is necessary to gently lower the gradient at points where performance using Adaptive Wing loss becomes stagnant. Therefore, we defined the parameters for Squared Adaptive Wing loss based on the values at these stagnation points. When using Adaptive Wing loss, the model's performance began to plateau between a loss of 0.0005 and 0.0006, beyond which no further improvements were seen, and the loss did not decrease further; in some cases, performance even worsened. At this point, the gradient of the Adaptive Wing loss was approximately 6.02, which is considerably high. Thus, using the value of 0.0005 for Adaptive Wing loss as a reference, we selected the ω and θ for Squared Adaptive Wing loss. To prevent the gradient from becoming too small prematurely, we set ω to 2,000 and θ to 0.2. Additionally, we adjusted the parameters ω' and θ' dynamically according to (5)–(6), allowing these values to adapt

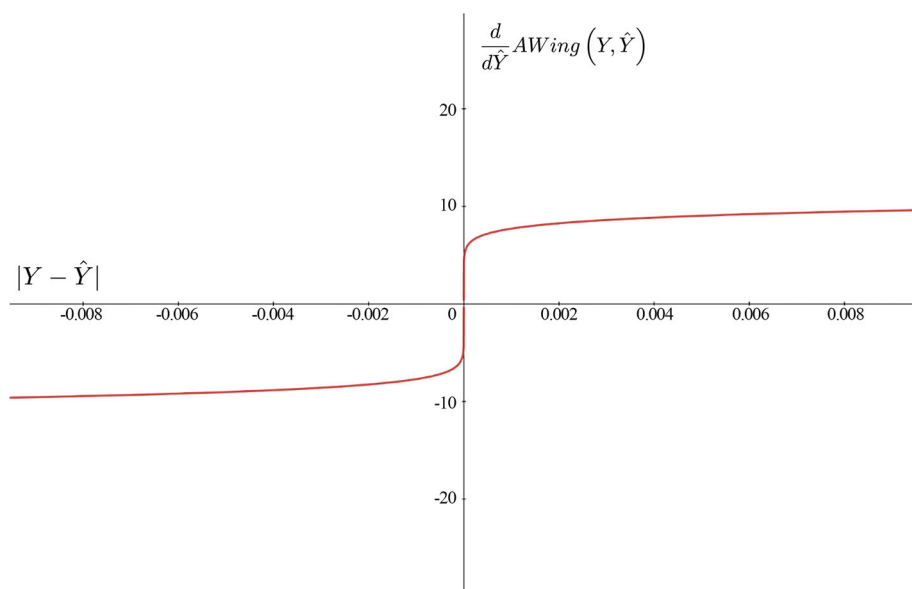


Fig. 4 The first derivative graph of the Adaptive Wing loss concerning $|Y - \hat{Y}|$ when Y equals to 1

based on Y . This adjustment ensures that the gradient decreases smoothly and consistently to zero, facilitating more precise weight convergence compared to the standard Adaptive Wing loss. For example, with a discrepancy of $|Y - \hat{Y}| = 0.0005$, the gradient stands at 4.81, which significantly reduces to 0.7 when the loss decreases to 0.0001.

$$\omega' = \omega + 9 \cdot \omega \cdot Y \quad (5)$$

$$\theta' = \theta - 0.99 \cdot \theta \cdot Y \quad (6)$$

$$A = \frac{\omega' \cdot (2\alpha - 2Y) \cdot \theta'^{2\alpha-2Y-1}}{1 + \theta'^{2\alpha-2Y}} \quad (7)$$

$$C = A \cdot \theta' - \omega' \cdot \log(1 + \theta'^{2\alpha-2Y}) \quad (8)$$

$$\text{Squared Adaptive Wing Loss}(Y, \hat{Y}) = \begin{cases} \omega' \cdot \log(1 + |Y - \hat{Y}|^{2\alpha-2Y}) & , |Y - \hat{Y}| < \theta' \\ A \cdot |Y - \hat{Y}| - C & , \text{otherwise} \end{cases} \quad (9)$$

The Soft-argmax function is pivotal in converting heatmap representations into coordinates, clearly distinguishing between the heatmap's foreground and background intensities. This method ensures accurate coordinate transformation as long as there is a significant difference between these intensities. Therefore, maintaining lower background values suffices for achieving precise results, without the need for meticulous learning of these background levels.

A major advantage of this approach, as depicted in Fig. 5, is that the gradient can be converged swiftly to zero when Y is 0, ensuring that the gradient remains very close to zero. This is particularly beneficial as it prevents any significant shifts in the model's weights due to high gradient values typical when using the Adaptive Wing loss, where the gradient may still be large even as background values approach zero. Conversely, in the case of using the Squared Adaptive wing, when Y is 1, although the gradient starts high, it is designed to decrease more gently as $|Y - \hat{Y}|$ approaches very close to 0, compared to the abrupt reductions seen with the Adaptive Wing loss shown in Fig. 4.

3.2 Heatmaps

As depicted in Fig. 6, it is essential to consider the distance between the rodent and several keypoints to accurately determine the specific arm of the Y-maze in which the experimental rodent is located. These points include the endpoints of arms A, B, and C and the center of the device. Therefore, it is crucial not only to track the position of the rodent but also to accurately learn the locations of each arm's endpoint and the device's center. This comprehensive understanding helps in effectively monitoring and analyzing the rodent's movements within the maze.

When creating a heatmap for arm keypoints using a Gaussian distribution, the result is a 2D representation as shown in Fig. 7, while its 3D form is depicted in Fig. 8. Notably, the end of each arm aligns with the center of the linear edge in these representations. To enhance performance, it is beneficial to design a heatmap that accurately reflects the edge shape of the arm ends. Towards this aim, we have introduced the Ellipse and Stadium heatmap configurations, which are specifically tailored to emphasize the consistent shape of the arm

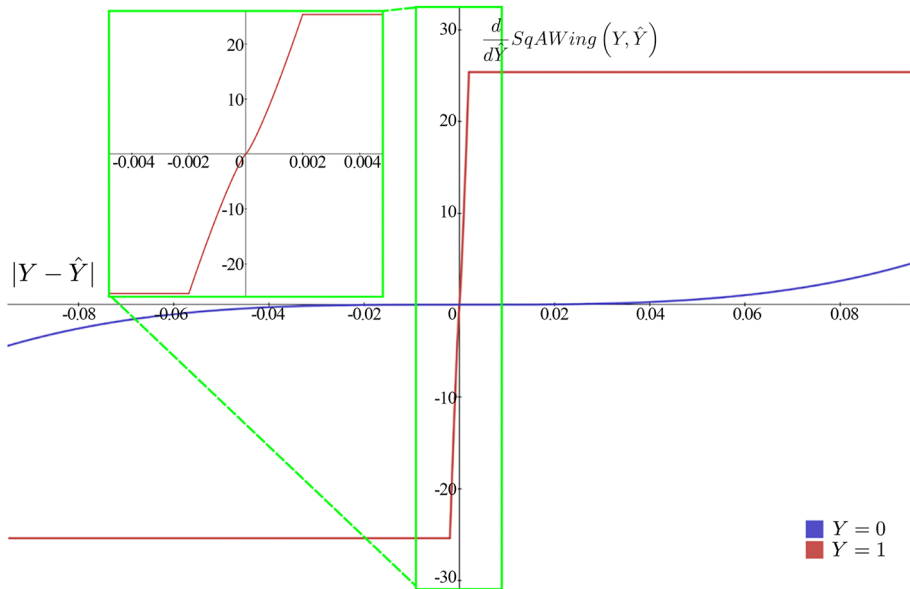


Fig. 5 The graphs depict the first derivatives of the Squared Adaptive Wing loss concerning $|Y - \hat{Y}|$, plotted separately for the cases when Y equals 1 and Y equals 0. In the top left inset, the graph corresponding to the case where Y equals 1 is magnified, specifically focusing on the region where $|Y - \hat{Y}|$ is very close to zero

ends, as cited in [10]. These configurations help more accurately track the arm endpoints, potentially improving the overall effectiveness of the keypoint detection.

3.2.1 Ellipse heatmap

To implement corner shapes in heatmaps effectively, the Ellipse Gaussian distribution can be employed. Ellipse heatmaps are widely used in applications such as Face Alignment and Pose Estimation to boost performance. For instance, in Face Alignment, these heatmaps are

Fig. 6 Analyzing the rodent's position using the distances between the rodent and each arm's end, as well as the center of the device

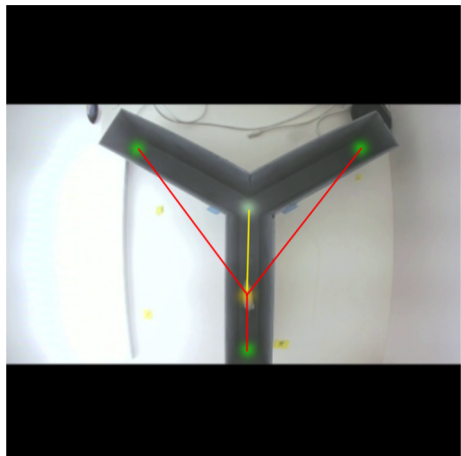
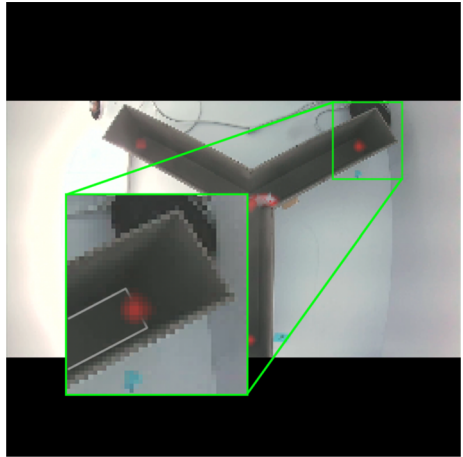


Fig. 7 The heatmap of an arm end based on general Gaussian distribution



crafted along the natural contours of facial features, including jawlines, eyebrow curves, and the outlines of lips. Similarly, in Pose Estimation, Ellipse heatmaps are created following the lines that connect joint keypoints.

We have developed a method specifically designed to compute Ellipse heatmaps for Y-Maze configurations. These heatmaps are elongated along the corner directions to represent better the unique shapes found in such setups. Figure 9 shows the 2D appearance of our proposed Ellipse heatmap, and Fig. 10 demonstrates its 3D form. These illustrations highlight how our specialized heatmaps adapt to the directional characteristics of corners, potentially enhancing the accuracy of our spatial analyses.

Constructing an Ellipse heatmap is systematically outlined through (10)–(14), which involve calculating the angles and lengths of each arm within the Y-Maze. Specifically, with the arm length set at 35cm, (10) is used to determine the scale factor by measuring the distance between the center point and the arm endpoints. (11) calculates the angles between the center point and each endpoint to position the arms in the heatmap accurately.

Further, (12) describes a matrix arrangement of coordinates along the Y and X axes, which is then rotated according to the calculated arm angles as specified in (13). Finally, (14) applies these dimensions and orientations to generate the heatmap Z using an Ellipse Gaussian distribution. This method ensures that the heatmap accurately reflects the geometric

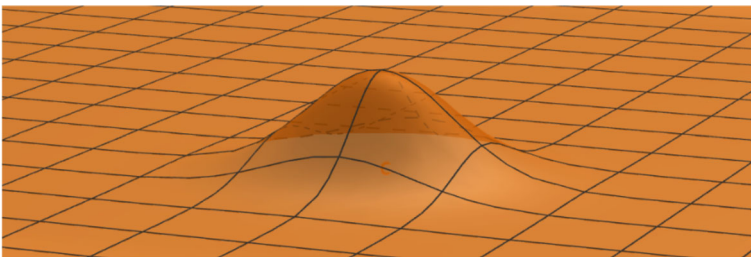
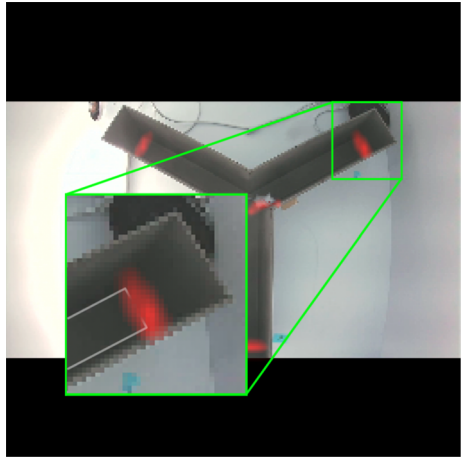


Fig. 8 3D graph of the general heatmap

Fig. 9 heatmap based on Ellipse Gaussian distribution



configuration of the arms, enhancing the precision of keypoint detection.

$$\rho = 35\sqrt{(x_{mid} - x_{arm})^2 + (y_{mid} - y_{arm})^2} \quad (10)$$

$$\theta = \tan^{-1} \left(\frac{x_{mid} - x_{arm}}{y_{mid} - y_{arm}} + \frac{\pi}{2} \right) \quad (11)$$

$$M = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ h & \cdots & h \end{pmatrix}, N = \begin{pmatrix} 0 & \cdots & w \\ \vdots & \ddots & \vdots \\ 0 & \cdots & w \end{pmatrix} \quad (12)$$

$$\begin{pmatrix} N' \\ M' \end{pmatrix} = (N - x_{arm} \ M - y_{arm}) \cdot \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \quad (13)$$

$$Z = e^{-\frac{(N' - x_{arm})^2}{2\sigma^2} - \frac{(M' - y_{arm})^2}{2(\sigma\rho)^2}} \quad (14)$$

3.2.2 Stadium heatmap

The Stadium heatmap, depicted in Fig. 11, can be constructed similarly to the Ellipse heatmap. Unlike traditional heatmap methods that require the neural network to identify a single maximum value precisely, the Stadium heatmap's design is more generous. Its maximum

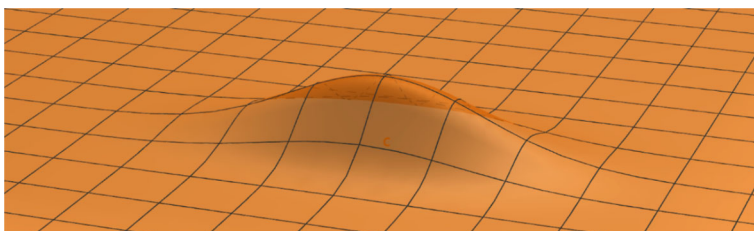
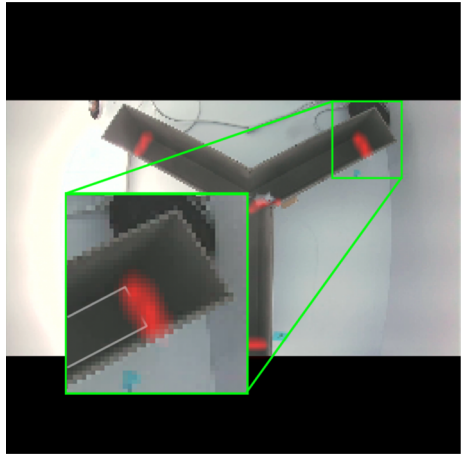


Fig. 10 3D graph of the Ellipse heatmap

Fig. 11 Heatmap based on Stadium Gaussian distribution



value extends across the entire length of the line, allowing for some inaccuracies in peak detection. This means that even if there are errors in predicting the exact highest point, the overall performance is not compromised as long as the majority of the foreground values are accurately identified.

A significant advantage of this approach is its ability to average out minor inaccuracies across the heatmap. The Stadium heatmap compensates for potential errors in maximum value prediction by considering the aggregate of correctly predicted points. Therefore, even if the Adaptive Wing loss leads to some deviations in individual weights from the optimal values, the overall system performance can still be improved or maintained. This averaging effect ensures that the impact of any single erroneous prediction is mitigated, making the Stadium heatmap a robust solution for consistent performance under varying conditions.

Equations (15) and (16) outline the formulas for constructing a Stadium heatmap. (15) modifies the axis center points to promote a uniform Gaussian distribution across the scale factor ρ , which is calculated in (10). Utilizing the matrix specified in (13), (16) is then used to produce the Stadium heatmap Z . This heatmap can be depicted in a 3D graphical format, similar to the stadium-shaped Gaussian distribution shown in Fig. 12. This visualization aids in understanding the spatial distribution and the adequate coverage of the heatmap.

$$\Psi(m) = \begin{cases} y_{arm} - \frac{\rho}{2} & (m \leq y_{arm} - \frac{\rho}{2}) \\ y_{arm} & (y_{arm} - \frac{\rho}{2} < m < y_{arm} + \frac{\rho}{2}) \\ y_{arm} + \frac{\rho}{2} & (y_{arm} + \frac{\rho}{2} \leq m) \end{cases} \quad (15)$$

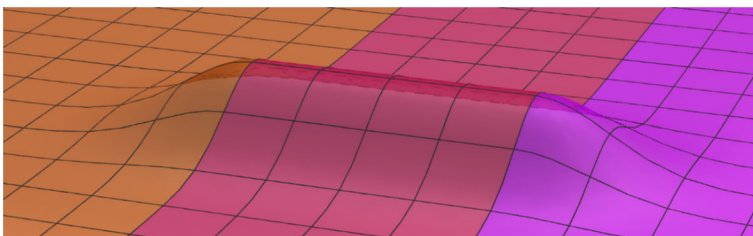


Fig. 12 3D graph of the Stadium heatmap

$$Z = e^{-\frac{(N' - x_{arm})^2 + (M' - \Psi(M))^2}{2\sigma^2}} \quad (16)$$

4 Implementation and evaluation

4.1 Dataset

The dataset for this experiment includes video recordings from the Y-Maze behavior test, which are used for behavioral analysis. As depicted in Fig. 13, the videos were captured from a top-down perspective, focusing on the center of the experimental device in alignment with the Y-Maze protocol. The dataset comprises 40,691 images in the training set, 5,155 images in the validation set, and 5,000 images in the test set. Each image was uniformly resized to a 256px by 256px square format by padding the edges with zeros to maintain consistency in data dimensions.

4.2 Image augmentation

To ensure that our deep learning models are trained without bias and have robust generalization capabilities, we have implemented image augmentation techniques extensively. These augmentations included adjusting contrast randomly between 50% and 150%, saturation levels from 50% to 150%, brightness from 70% to 130%, and JPEG quality from 20% to 100%. Additionally, we used random rotations between -180° and 180° , translations from -5% to 5%, and channel shifts with a 50% probability per channel. To further enhance the dataset's variability, we randomly applied grayscale conversion, intensity reversal, and cutouts affecting 30% of the area with a 50% probability each. These comprehensive measures help in creating a more balanced and representative training environment.

4.3 Metric

In the realm of keypoint detection, the Normalized Mean Error (NME) metric, defined in (17), plays a crucial role in ensuring that the deep learning model is accurately aligned

Fig. 13 A sample image of the dataset



with its training objectives. The NME metric measures the Euclidean distance but adjusts for the varying scales of subjects across different datasets. This adjustment is essential for meaningful comparisons.

As illustrated in Fig. 14, comparing the absolute differences $|E_1 - E_2|$ and $|E_3 - E_4|$ between two images directly is not practical due to scale discrepancies. To overcome this issue, we use a scaling factor d that represents the subject's scale in each image. This factor normalizes the differences or errors, thereby allowing for a valid comparison of keypoints across images of different sizes and scales.

The failure rate is a critical metric for evaluating the performance of facial landmark detection models, representing the proportion of instances where the model's error exceeds a specified threshold. Generally, this threshold defines the acceptable range of landmark coordinate errors, with any instance exceeding this threshold considered a failure. The failure rate is calculated as the percentage of test samples classified as failures. For example, if the threshold is set at 0.1, samples with a Normalized Mean Error (NME) between predicted and actual landmark coordinates exceeding 0.1 are recorded as failures. This metric is particularly useful for assessing how robustly the model performs under extreme conditions, providing insights into its stability and reliability. In this study, we measured the failure rate using a threshold based on 1.14% of the arm length (35 cm), corresponding to 4 mm. Thus, instances where the NME between predicted and actual landmark coordinates exceeded 4 mm were considered failures. This stringent criterion allows for a more rigorous and realistic evaluation of the model's performance. When used alongside NME, the failure rate offers a comprehensive assessment of both the overall performance and the model's ability to handle exceptional cases.

$$\text{NME}(\%) (Y, \hat{Y}) = \frac{100}{n} \sum_i^n \frac{\|Y_i - \hat{Y}_i\|_2}{d} \quad (17)$$

As illustrated in Fig. 15, the scale factor d , which represents the scale of the subject, is determined by the distance between the Y-maze's center point and the arm end. This scale factor is crucial for standardizing measurements across different setups and is clearly defined in the experimental rules. The value of d can vary, depending on the arm length as measured from the images provided in the experiment. Using d as a consistent measure, we

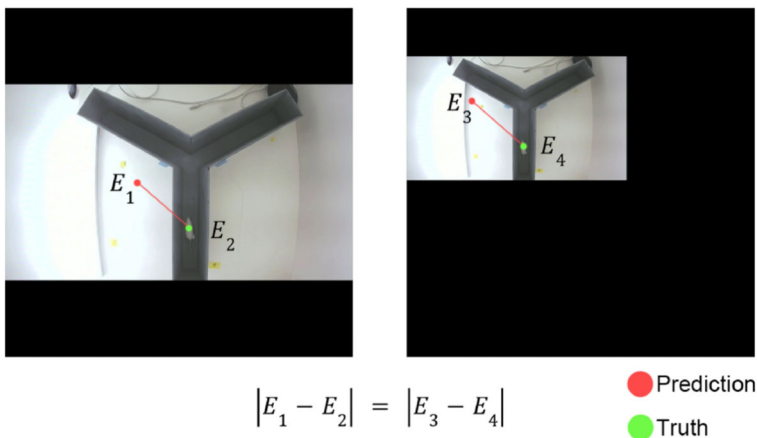


Fig. 14 Different scale images



Fig. 15 The arm length d for Y-maze

can accurately compare results from different Y-maze configurations and ensure that the data are normalized to reflect true differences in subject behavior.

4.4 Neural network training

Neural network training was performed over 80 epochs, with the only variables being the type of loss function and the heatmap configuration used for the endpoints of arms A, B, and C. We employed the Adam optimizer for this process, with an initial learning rate of 0.001. This rate was subject to exponential decay, reducing by a factor of 0.96 every four epochs to optimize performance gradually.

The architecture of the neural network was based on a four-stacked HourglassNet, which included four distinct Hourglass Modules. To facilitate effective training, the network was initially equipped with two Residual Modules and a max pooling layer, configured to achieve a heatmap resolution of 64×64 .

The experiments were conducted on a high-performance server computer equipped with an AMD Ryzen ThreadRipper PRO 3955WX CPU, 192GB of RAM, and four Nvidia A5000 GPUs. This setup allowed for efficient data processing and model training. Specifically, the server was capable of training at a rate of 14.1 data samples per second and processing 39.9 data samples per second during inference. This robust computational environment ensured that our models could be trained and evaluated effectively, providing reliable performance metrics for our study.

5 Experimental results

5.1 Experiments with adaptive wing loss

Table 1 presents a comparative analysis of the average Normalized Mean Error (NME) values across different heatmap configurations when trained using the Adaptive Wing loss function. This result reflects the performance at the endpoints of arms A, B, and C, utilizing three distinct types of heatmaps. Utilizing the four-stacked HourglassNet, which produces four outputs corresponding to the network's depth as illustrated in Fig. 3, it is observed that the final output consistently shows the best NME values.

Table 1 Average performance for each output when applying three heatmap methods to arm ends and adaptive wing loss used for training

Loss	Arm Heatmap	NME(%)				Failure Rate(%)			
		Out1	Out2	Out3	Out4	Out1	Out2	Out3	Out4
AWing	General	0.94	0.88	0.88	0.88	16.22	9.94	9.97	10.43
	Ellipse	0.96	0.85	0.84	0.84	20.07	8.22	7.76	8.12
	Stadium	0.95	0.86	0.86	0.85	21.97	9.84	8.85	8.69

Values are averages of 5 experiments

Specifically, as shown in Table 1, the average NME value for the final output from the four-stacked HourglassNet is 0.88% when using the general Gaussian heatmap, 0.84% with the Ellipse heatmap, and 0.85% with the Stadium heatmap. These results highlight that both the Ellipse and the newly proposed Stadium heatmap significantly enhance the model's overall performance, even though these are only applied to the endpoints of arms A, B, and C. Therefore, it is essential to define the heatmap according to the shape of the subject for effective learning. The failure rate was also the lowest when using the ellipse heatmap, registering at 8.12%. This demonstrates the superior performance of the ellipse heatmap method in our facial landmark detection experiments.

Table 2 offers a detailed comparison of the NME values for each keypoint, highlighting the impact of different heatmap types applied to the endpoints of arms A, B, and C. The general Gaussian heatmap, typically used for keypoints other than the arm endpoints, shows similar NME values across these other keypoints, indicating a consistent performance regardless of the heatmap type.

However, distinct differences in performance are observed at the arm endpoints based on the heatmap applied. With the general Gaussian heatmap, the NME for Output 4 is recorded at 0.73%, 0.69%, and 0.87% for the endpoints of arms A, B, and C. In contrast, they used the Stadium heatmap results in significantly improved prediction accuracy with NME values of 0.72%, 0.62%, and 0.71%, respectively. The Ellipse heatmap further enhances performance, achieving NME values of 0.69%, 0.56%, and 0.72% for the respective arms. These results clearly demonstrate the superior efficacy of the Stadium and Ellipse heatmaps in improving keypoint detection at the arm endpoints.

The provided graph in Fig. 16 offers a visual comparison of the Normalized Mean Error (NME) for each arm of the Y-Maze when trained with the Adaptive Wing loss function, utilizing General, Ellipse, and Stadium heatmap approaches.

As the graph demonstrates, the Ellipse heatmap yields the most notable performance improvement across the arm keypoints. When comparing the different heatmaps, it's evident that while the General heatmap offers consistent results, the NME for arm B (shown in green) decreases significantly with the Ellipse heatmap, reaching the lowest value of 0.56%. This represents a substantial improvement over the General heatmap's 0.69% NME. Arm C (in blue) also shows a marked improvement with the Ellipse heatmap, matching the performance of the General heatmap at 0.72%.

In contrast, the Stadium heatmap appears to provide mixed results. While it improves the NME for arm B to 0.62%, surpassing the General heatmap, it does not achieve the same level of performance enhancement as seen with the Ellipse heatmap. Additionally, the performance for arm A (in orange) with the Stadium heatmap remains the same as with the General heatmap, indicating that the Stadium heatmap does not universally improve NME across all keypoints.

Table 2 Performance for each keypoint with the application of Adaptive Wing loss and three heatmap methods to arm ends

Arm Heatmap	Keypoints	NME(%)				Failure Rate(%)			
		Out1	Out2	Out3	Out4	Out1	Out2	Out3	Out4
General	Arm A end	0.76	0.74	0.73	0.73	33.39	28.02	27.52	27.44
	Arm B end	0.72	0.68	0.68	0.69	14.22	9.68	10.65	11.32
	Arm C end	0.93	0.89	0.88	0.87	31.36	25.46	25.88	26.86
	Center of device	1.02	0.95	0.95	0.95	13.79	10.32	10.09	9.54
	Body of mouse	1.01	0.95	0.94	0.96	30.24	28.39	27.91	29.34
	Head of mouse	1.12	1.06	1.06	1.06	40.33	34.68	35.00	35.04
	Tail of mouse	1.00	0.92	0.91	0.92	31.15	27.07	26.64	27.21
Ellipse	Arm A end	0.77	0.70	0.69	0.69	35.83	24.76	23.30	24.48
	Arm B end	0.68	0.59	0.56	0.56	8.57	5.76	3.99	4.11
	Arm C end	0.86	0.73	0.73	0.72	19.83	11.89	12.50	12.13
	Center of device	1.07	0.91	0.89	0.89	20.76	10.06	8.57	8.83
	Body of mouse	1.06	0.98	0.98	0.99	36.14	31.47	32.07	32.11
	Head of mouse	1.19	1.08	1.08	1.07	45.83	36.42	36.81	36.58
	Tail of mouse	1.07	0.94	0.94	0.95	36.25	29.33	29.74	29.44
Stadium	Arm A end	0.78	0.72	0.72	0.72	34.72	26.64	22.58	22.14
	Arm B end	0.72	0.62	0.62	0.62	17.42	10.58	8.94	7.44
	Arm C end	0.78	0.72	0.70	0.71	27.85	14.1	13.52	13.43
	Center of device	1.04	0.94	0.94	0.94	15.02	9.68	9.01	9.01
	Body of mouse	1.09	1.00	1.00	0.98	35.82	31.68	32.72	34.45
	Head of mouse	1.18	1.09	1.08	1.07	44.72	37.23	37.31	37.39
	Tail of mouse	1.06	0.94	0.95	0.93	36.87	27.31	26.38	26.17

Values are averages of 5 experiments

Overall, these results suggest that the choice of heatmap significantly impacts the model's accuracy.

5.2 Experiments with squared adaptive wing loss

The comparative effectiveness of the Adaptive Wing loss and the Squared Adaptive Wing loss is highlighted in the experimental results shown in Tables 1 and 3, respectively. These experiments involved applying three distinct types of heatmaps to the endpoints of arms A, B, and C, while general Gaussian heatmaps were used for other keypoints.

When using the general Gaussian heatmap at the arm endpoints and considering Output 4 as the benchmark, the Adaptive Wing loss resulted in an NME of 0.88%. In contrast, the Squared Adaptive Wing loss improved this figure to 0.77%, marking a 0.1% enhancement in performance. With the Ellipse heatmap, the NME under Adaptive Wing loss was 0.84%, compared to a significantly better 0.74% with Squared Adaptive Wing loss. Notably, the Squared Adaptive Wing loss also outperformed the Adaptive Wing loss when using the Stadium heatmap, achieving an NME of 0.72% against 0.85%. This data underscores the performance gains possible with the Squared Adaptive Wing loss across different heatmap configurations, as detailed in Section 5.1. The failure rate was the lowest when using the

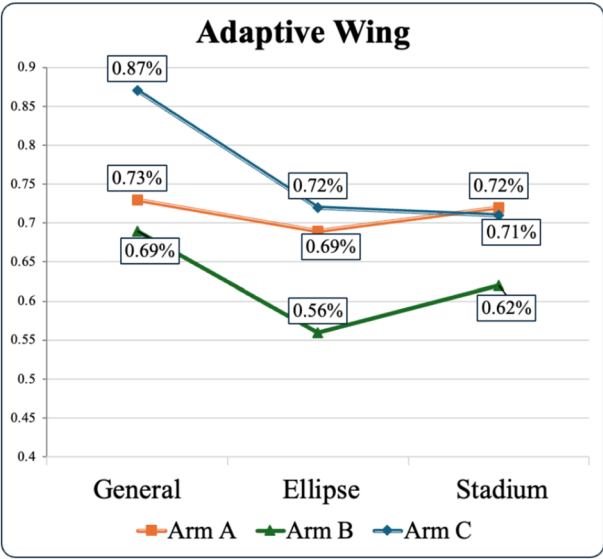


Fig. 16 Comparative graph of keypoint detection accuracy across heatmap configurations under Adaptive Wing loss

Stadium heatmap, achieving a rate of 2.46%. This represents a significant improvement of 5.66 percentage points over the performance with Adaptive Wing Loss, which had a failure rate of 8.12%. This demonstrates the superior performance of the Stadium heatmap method in our facial landmark detection experiments.

The graph in Fig. 17 illustrates the Arm ends prediction efficacy of two different loss functions-Adaptive Wing (AWing) and Squared Adaptive Wing (SqAWing)—across three heatmap configurations: General, Ellipse, and Stadium. In this visual comparison, it is apparent that the Squared Adaptive Wing loss consistently outperforms the Adaptive Wing loss across all heatmap types.

For the General heatmap, the NME decreases from 0.76 with AWing to 0.69 with SqAWing, indicating a substantial improvement. When the Ellipse heatmap is applied, the improvement is even more pronounced, with the NME dropping from 0.66 under AWing to 0.57 with SqAWing. This trend continues with the Stadium heatmap, where both loss functions achieve an NME of 0.57, signifying that SqAWing matches the best performance of AWing even in this configuration.

Table 3 Average performance for each output when applying three heatmap methods to arm ends and Squared Adaptive Wing loss used for training

Loss	Arm Heatmap	NME(%)				Failure Rate(%)			
		Out1	Out2	Out3	Out4	Out1	Out2	Out3	Out4
Sq. AWing	General	0.90	0.78	0.77	0.77	11.19	4.52	3.94	4.04
	Ellipse	0.89	0.75	0.74	0.74	10.86	3.99	3.73	3.72
	Stadium	0.92	0.74	0.73	0.72	11.70	2.70	2.69	2.46

Values are averages of 5 experiments

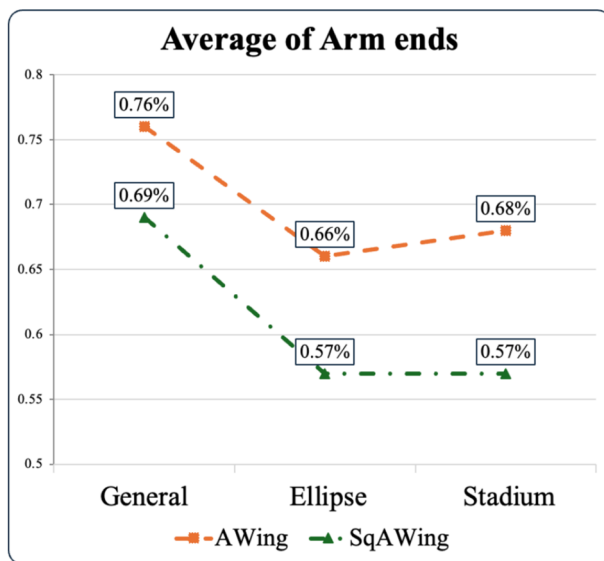


Fig. 17 Average performance comparison of loss functions across heatmap types for Y-Maze arm ends detection

Overall, the data portrayed in this graph suggest a consistent advantage when utilizing Squared Adaptive Wing loss in terms of average NME across different heatmap types. This finding is particularly relevant for improving the precision of keypoint detection algorithms in behavioral tests such as the Y-Maze. It confirms that the Squared Adaptive Wing loss can provide a more robust and accurate model for such applications.

Table 4 showcases the keypoint-specific results when utilizing the Squared Adaptive Wing loss function and applying three distinct heatmap types to the arms A, B, and C endpoints. The data indicate that there was not only an improvement in the NME for the arm endpoints but also a noticeable enhancement across other keypoints.

Employing the general Gaussian heatmap at the arm endpoints significantly improved the performance for the B arm, achieving an NME of 0.63%, which is an improvement from the 0.69% NME observed with the Adaptive Wing loss. Moreover, using the Ellipse heatmap further enhanced the NME to 0.50%. The Stadium heatmap produced the best results among the three, with an NME of 0.47%, surpassing even the Ellipse heatmap's performance. However, it is important to note that there was a slight decline in performance for the endpoint of arm A when using the Stadium heatmap.

Overall, the Squared Adaptive Wing loss typically shows superior predictive performance compared to the Adaptive Wing loss. Notably, combining the Squared Adaptive Wing loss with the Ellipse heatmap for the arm endpoints offers the most optimal performance, reflecting its effectiveness in refining keypoint detection accuracy.

The graph in Fig. 18 displays the Normalized Mean Error (NME) for each arm of the Y-Maze, with respect to General, Ellipse, and Stadium heatmap types. What stands out is that the performance is quite comparable between the Ellipse and Stadium heatmaps, but with a lower variance in the results of the Stadium heatmap, suggesting it offers more stable performance across different keypoints.

Table 4 Performance for each keypoint with applying Squared Adaptive Wing loss and three heatmap methods to arm ends

Arm Heatmap	Keypoints	NME(%)				Failure Rate(%)			
		Out1	Out2	Out3	Out4	Out1	Out2	Out3	Out4
General	Arm A end	0.77	0.66	0.66	0.66	25.56	14.66	14.01	14.15
	Arm B end	0.75	0.63	0.62	0.63	13.03	4.54	3.89	3.51
	Arm C end	0.91	0.79	0.77	0.78	27.64	19.84	17.64	17.55
	Center of device	0.93	0.75	0.74	0.73	12.76	6.35	5.42	5.47
	Body of mouse	0.92	0.87	0.86	0.86	27.01	22.77	22.22	22.37
	Head of mouse	1.06	0.93	0.92	0.92	37.61	29.31	27.49	27.70
	Tail of mouse	0.92	0.84	0.83	0.83	27.36	19.11	18.73	19.56
Ellipse	Arm A end	0.77	0.66	0.66	0.66	29.29	22.33	21.86	20.78
	Arm B end	0.64	0.52	0.50	0.50	8.03	3.01	2.15	2.57
	Arm C end	0.77	0.55	0.55	0.55	15.45	4.66	3.86	4.01
	Center of device	1.05	0.80	0.79	0.79	14.16	7.28	7.70	8.16
	Body of mouse	0.96	0.90	0.89	0.88	32.63	28.99	26.99	26.67
	Head of mouse	1.09	0.95	0.95	0.95	39.14	30.17	29.69	29.27
	Tail of mouse	0.95	0.86	0.85	0.85	27.35	22.96	22.33	21.95
Stadium	Arm A end	0.86	0.70	0.71	0.69	31.39	14.91	14.75	15.4
	Arm B end	0.71	0.50	0.46	0.47	6.75	2.35	1.94	2.28
	Arm C end	0.89	0.60	0.56	0.55	17.06	6.88	6.58	4.65
	Center of device	0.98	0.72	0.71	0.71	15.65	6.50	4.37	3.60
	Body of mouse	0.97	0.88	0.88	0.87	29.46	23.89	23.53	23.30
	Head of mouse	1.08	0.94	0.92	0.92	37.26	28.26	26.51	26.88
	Tail of mouse	0.96	0.86	0.84	0.83	27.4	20.68	20.39	19.94

Values are averages of 5 experiments

Specifically, when utilizing the General heatmap, the NME for Arm C is 0.78%, which is the highest among the results. The Ellipse heatmap and Stadium heatmap lead to an NME of 0.55% for the same keypoint, showing an improvement. The most significant observation is for Arm C, where the Stadium heatmap has an NME of 0.47%, slightly lower than the Ellipse heatmap's 0.50%, showing the highest performance gain across all heatmap types and keypoints.

In conclusion, while both the Ellipse and Stadium heatmaps enhance keypoint detection performance when trained with the Squared Adaptive Wing loss function, the Stadium heatmap's performance is slightly more consistent, indicating it might be a more reliable choice for models where stability across various keypoints is essential.

5.2.1 Stability comparison: NME trends of two loss functions

Figure 19 displays a graph that tracks the progression of Normalized Mean Error (NME) changes for the validation set across different epochs when trained using both Adaptive Wing loss and Squared Adaptive Wing loss. The NME trend for the models trained with Adaptive Wing loss exhibits a somewhat erratic pattern, significantly rising and stabilizing periodically. In contrast, the trend for models trained with Squared Adaptive Wing loss

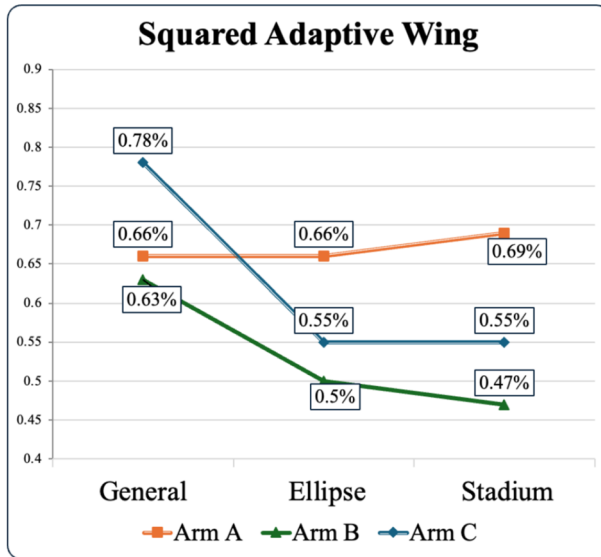


Fig. 18 Arm ends prediction performance comparison using squared adaptive wing loss

shows a more consistent decrease in NME, without the marked fluctuations observed with the Adaptive Wing loss. This indicates a steadier and potentially more reliable improvement in model accuracy over epochs when using Squared Adaptive Wing loss.

6 Conclusions

In this study, we addressed the challenges of automating the Y-maze behavioral test, a commonly used method to assess short-term spatial memory in rodents. To address the limitations of continuous human monitoring, we proposed an automated system that leverages keypoint detection technology for precisely tracking rodents and the Y-Maze apparatus. The computer

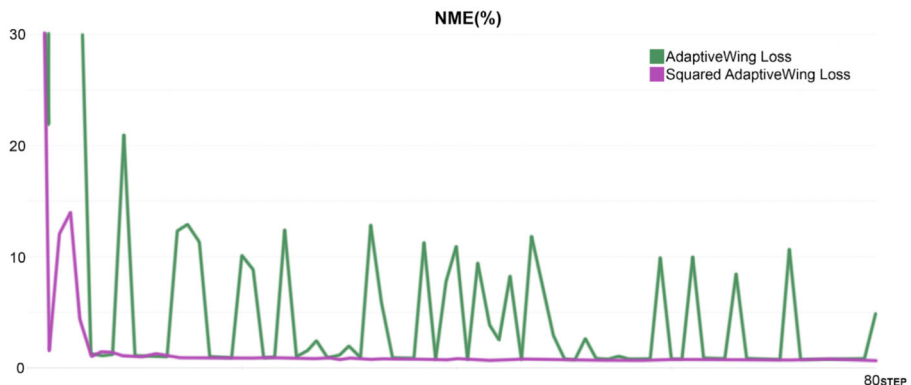


Fig. 19 Graph of validation NMEs for adaptive wing and squared adaptive wing

vision deep learning method for the automation process detailed in this study facilitates high throughput analysis, allowing for the evaluation of many subjects simultaneously without compromising data quality. This efficiency is vital in fields where time is of the essence, such as drug testing and neurological research.

Specifically, the Stadium heatmap and the Squared Adaptive Wing loss have significantly improved prediction accuracy, as evidenced by the reduction in Normalized Mean Error (NME). These advancements facilitate the automation of the Y-Maze test. In addition, While the study focuses on the Y-maze test, the methodologies proposed have broader implications. The heatmaps and loss functions can be adapted for other domains that rely on keypoint detection, such as human pose estimation and gait analysis, and even in broader contexts within computer vision and machine learning.

Looking ahead, further research should investigate the potential of these novel techniques in other areas of keypoint detection, such as Pose Estimation and Face Alignment, to broaden their applicability and impact.

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Data Availability The data is not publicly available at the request of the provider, but is depicted in detail in Figs. 13–15.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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References

1. Bannerman DM, Niewoehner B, Lyon L, Romberg C, Schmitt W.B, Taylor A, Sanderson DJ, Cottam J, Sprengel R, Seeburg PH, Köhr G, Rawlins JNP (2008) Nmda Receptor Subunit NR2A Is Required for Rapidly Acquired Spatial Working Memory But Not Incremental Spatial Reference Memory. *J Neurosci* 28(14):3623–3630. [Online; accessed 6 Sep 2023]
2. Y Maze Spontaneous Alternation Test. <https://med.stanford.edu/sbfnl/services/bm/lm/y-maze.html>. [Online; accessed 6 Sep 2023]
3. Oh E, Kang J-H, Jo KW, Shin W-S, Jeong Y.H, Kang B, Rho T-Y, Jeon SY, Lee J, Song I-S, Kim K-T (2022) Synthetic PPAR Agonist DTMB Alleviates Alzheimer's Disease Pathology by Inhibition of Chronic Microglial Inflammation in 5xfad Mice. *Neurotherapeutics* 19(5):1546–1565. [Online; accessed 6 Sep 2023]
4. Liu H, Liu F, Fan X, Huang D (2021) Polarized self-attention: towards high-quality pixel-wise regression. [arXiv:2107.00782](https://arxiv.org/abs/2107.00782). [Online; accessed 6 Sep 2023]
5. Zhang J, Chen Z, Tao D (2020) Towards high performance human keypoint detection. [arXiv:2002.00537v2](https://arxiv.org/abs/2002.00537v2). [Online; accessed 6 Sep 2023]

6. Wang J, Sun K, Cheng T, Jiang B, Deng C, Zhao Y, Liu D, Mu Y, Tan M, Wang X, Liu W, Xiao B (2019) Deep high-resolution representation learning for visual recognition. [arXiv:1908.07919v2](#). [Online; accessed 6 Sep 2023]
7. Artacho B, Savakis A (2021) OmniPose: a multi-scale framework for multi-person pose estimation. [arXiv:2103.10180v1](#). [Online; accessed 6 Sep 2023]
8. Zhou Z, Li H, Liu H, Wang N, Yu G, Ji R (2023) STAR loss: reducing semantic ambiguity in facial landmark detection. [arXiv:2306.02763v1](#). [Online; accessed 6 Sep 2023]
9. Lan X, Hu Q, Chen Q, Xue J, Cheng J (2021) HIH: towards more accurate face alignment via heatmap in heatmap. [arXiv:2104.03100v2](#). [Online; accessed 6 Sep 2023]
10. Huang Y, Yang H, Li C, Kim J, Wei F ICCV 2021 Open access repository. https://openaccess.thecvf.com/content/ICCV2021/html/Huang_ADNet_Leveraging_Error-Bias_Towards_Normal_Direction_in_Face_Alignment_ICCV_2021_paper.html. [Online; accessed 6 Sep 2023]
11. Wang X, Bo L, Fuxin L ICCV 2019 Open access repository. https://openaccess.thecvf.com/content_ICCV_2019/html/Wang_Adaptive_Wing_Loss_for_Robust_Face_Alignment_via_Heatmap_Regression_ICCV_2019_paper.html. [Online; accessed 6 Sep 2023]
12. Huang J, Zhu Z, Guo F, Huang G, Du D (2019) The devil is in the details: delving into unbiased data processing for human pose estimation. [arXiv:1911.07524](#). [Online; accessed 6 Sep 2023]
13. Xu Z, Li B, Geng M, Yuan Y (2020) AnchorFace: an anchor-based facial landmark detector across large poses. [arXiv:2007.03221v3](#). [Online; accessed 6 Sep 2023]
14. Luvizon D.C, Tabia H, Picard D (2017) Human pose regression by combining indirect part detection and contextual information. [arXiv:1710.02322](#). [Online; accessed 6 Sep 2023]
15. Newell A, Yang K, Deng J (2016) Stacked hourglass networks for human pose estimation. [arXiv:1603.06937v2](#). [Online; accessed 6 Sep 2023]
16. Ronneberger O, Fischer P, Brox T (2015) U-Net: convolutional networks for biomedical image segmentation. [arXiv:1505.04597](#). [Online; accessed 6 Sep 2023]
17. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Erhan D, Vanhoucke V, Rabinovich A (2014) Going deeper with convolutions. [arXiv:1409.4842](#). [Online; accessed 6 Sep 2023]

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